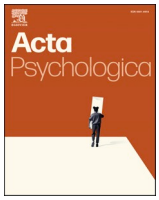
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The effectiveness and feasibility of the new Forest parenting programme for Chinese children with attention-deficit/hyperactivity disorder and their families

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ABSTRACT

Background: The New Forest Parenting Programme (NFPP) is a therapist-delivered parent training intervention aimed at improving behavioral difficulties in children with attention-deficit/hyperactivity disorder (ADHD) by promoting positive parenting practices. Although NFPP has shown effectiveness in several Western countries, evidence regarding its feasibility and effectiveness in the Chinese context remains limited. This study aims to evaluate its feasibility and effectiveness in China.

Methods: Thirty Chinese children with ADHD and their families were enrolled and sequentially allocated to the NFPP intervention group (n = 15) or control group (n = 15). Parent-report measures, including the Chinese Version of Swanson Nolan and Pelham, Version IV Scale-parent Form (SNAP-IV), Chinese Parenting Sense of Competence (C-PSOC), and Parenting Styles Dimensions Questionnaire (PSDQ), were used to assess quantitative outcomes and analyzed using analysis of covariance (ANCOVA). Finally, semi-structured interviews were conducted with the intervention group after completing the NFPP, and the qualitative data were analyzed using thematic analysis.

Results: Based on parent-report, compared with the control group, the intervention group showed significant reductions in inattention, hyperactivity/impulsivity, and oppositional defiant scores after the NFPP intervention. Parents reported increased parenting efficacy and satisfaction, along with increased use of an authoritative parenting style. The qualitative analysis offered insights into driving factors of NFPP's effectiveness, participant motivation, adherence barriers, limitations, and acceptability.

Conclusions: This non-randomized controlled trial suggests that NFPP is effective and feasible for Chinese children with ADHD and their families. Further validation in larger-scale randomized controlled trials is warranted.

1. Introduction

Attention-deficit/hyperactivity disorder (ADHD) is one of the most common psychiatric disorders in children. It's characterized by persistent and developmentally inappropriate symptoms of inattention and/or hyperactivity/impulsivity (American Psychiatric Association, 2013). The worldwide prevalence of ADHD is 5.29% (Polanczyk et al., 2007), while epidemiological studies report a higher prevalence among children in China (6.4%) (Li, Cui, et al., 2022). Children with ADHD often experience significant impairments in academic performance, social functioning, and executive processes such as working memory and organizational planning. Additionally, families of children with ADHD

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often face many challenges, including elevated levels of parenting stress, frequent family conflicts, greater reliance on harsh or coercive discipline, and less effective parenting practices (Johnston & Mash, 2001). Research suggests that these adverse parenting and family environments influence ADHD symptom expression and may be associated with an increased likelihood of an ADHD diagnosis (Claussen et al., 2024).

Stimulants are recommended as the first-line medication for treating ADHD. However, medication adherence is often suboptimal, with adverse effects and perceived lack of efficacy being among the most frequently reported reasons for discontinuation (Ferrin et al., 2025; Gajria et al., 2014). Compared to medication treatment, parents seem to have a higher acceptance of psychosocial interventions (Adler & Nierenberg, 2010; Shen et al., 2021). In addition, all clinical practice guidelines also emphasize the importance of non-pharmacological interventions (Coghill et al., 2023). Notably, parent training is recommended as first-line treatment for preschool-aged children with ADHD (Wolraich et al., 2019). In China, evidence-based parent training programs remain scarce. Some research groups have found that non-pharmacological interventions such as Integrated Brain, Body, and Social intervention and Executive Function Training have shown only limited effectiveness in improving ADHD symptoms (Shuai et al., 2021; Smith et al., 2020). A recent randomized controlled trial conducted in China demonstrated that a mindfulness-based stress reduction (MBSR) programme effectively alleviated stress levels among parents of children with ADHD; however, it did not produce significant improvements in the children's ADHD symptoms (Law et al., 2025). Although a parent education program conducted in Hong Kong presented positive effects on preschool children with developmental disabilities, including ADHD (Leung et al., 2016), it was not specifically designed as ADHD-focused parent training. Similarly, a Multiple Family Therapy for ADHD families conducted in China failed to show significant treatment effects (Ma et al., 2018). In summary, the interventions currently available for ADHD in China are not empirically supported. Therefore, there is an urgent need to introduce evidence-based, ADHD-specific parent training programs in the Chinese context.

The New Forest Parenting Programme (NFPP) originated in the United Kingdom and is a well-established, evidence-based parent training intervention (Russell et al., 2022; Thompson et al., 2009). Generic parenting programs such as Incredible Years (Webster-Stratton et al., 2008) and Triple P (Nowak & Heinrichs, 2008) are based on universal behavior modification principles. They are primarily effective in reducing oppositional behaviors with limited impact on core ADHD symptoms, the NFPP was specifically developed for children with ADHD. Drawing on key aetiological theories of ADHD, the NFPP directly targets ADHD-related behavioral characteristics and underlying psychological deficits, particularly impairments in executive functioning, with the aim of achieving sustained improvements in core symptoms across multiple settings (Sonuga-Barke et al., 2008; Thompson et al., 2009). Building on traditional behavior management strategies, the NFPP further incorporates interventions targeting neurocognitive deficits and, through parents' use of constructive parenting, ultimately promotes the development of self-control and self-regulation in children with ADHD. Randomized controlled trials in multiple countries, including the UK, the United States, and Denmark, have demonstrated that NFPP significantly improves ADHD symptoms, providing robust evidence of its efficacy (Abikoff et al., 2015; Daley & O'Brien, 2013; Häge et al., 2018; Lambek et al., 2023; Sonuga-Barke et al., 2001; Thompson et al., 2009).

In recent years, the NFPP has been introduced to China. It is primarily delivered as an individual therapist-led intervention. However, the high costs associated with therapist training and ongoing supervision, together with practical barriers such as parents' work commitments, may limit participation. To improve accessibility and reduce barriers to engagement, a self-help version of the NFPP (NFPP-SH) based on self-help materials has been proposed as an alternative approach. To date, only one study in China has evaluated NFPP-SH (Thompson et al., 2017). The NFPP-SH study targeted both parents and grandparents of children with ADHD. Although NFPP-SH was generally well received, its implementation encountered challenges, as grandparents experienced difficulties in fully understanding and consistently applying the recommended parenting strategies, which may have constrained the overall effectiveness of the intervention. Accordingly, the authors highlighted the potential value of therapist-led NFPP interventions in facilitating caregivers' understanding, adaptation, and implementation of the intervention strategies. Given the limited evidence for NFPP-SH in China and the potential challenges faced by caregivers in comprehending and applying intervention strategies, we therefore conducted a therapist-led NFPP intervention. To our knowledge, no study has systematically evaluated the therapist-led NFPP in Chinese families using both quantitative and qualitative approaches. Therefore, we conduct a small therapist-led trial to objectively assess its effectiveness using quantitative data. In addition, because NFPP was originally developed in the United Kingdom, its feasibility in the Chinese context remains uncertain. To address this, we conduct semi-structured interviews to further explore its feasibility by examining the following research questions: (a) the driving factor of NFPP's effectiveness; (b) participation motivations; (c) adherence barriers; (d) program limitations; (e) acceptability.

2. Methods

2.1. Study design

The participants in this study were recruited from the outpatient clinics of the hospital. Informed consent was obtained for all parents and children, and all consent forms were signed by the parents. The reporting of the quantitative research part of this study was in accordance with the Transparent Reporting of Evaluations with Non-randomized Designs, while the reporting of the qualitative part adhered to the Consolidated Criteria for Reporting Qualitative Research.

In this study, we employed quantitative and qualitative approaches. This approach enables a more thorough evaluation of an intervention (Campbell et al., 2000). Fig. 1 shows a conceptual diagram outlining the framework of our study.

2.2. Participants

Children included in this study met the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) criteria for ADHD, were under 12 years of age, and had primary caregivers who were able to understand and complete the relevant questionnaires. Families that met these criteria were included in the study. To ensure sample homogeneity and minimize potential confounding factors, families were excluded if any of the following applied to the child: (a) current use of ADHD medications or engagement in behavioral therapy; (b) comorbid intellectual disability; (c) comorbid other psychiatric disorders; (d) comorbid neurological disorders, organic brain diseases, or other neurodevelopmental disorders (such as autism spectrum disorder). During the study period, no children with ADHD received medication treatment.

We initially obtained informed consent from 32 families. However, two families met the exclusion criteria—one child had comorbid tic disorder, and the other was currently taking ADHD medication. Therefore, a total of 30 families of children with ADHD were included, with one parent from each family serving as a participant in this study. At baseline, information on the gender, age, and educational level of both the children with ADHD and their parents was collected.

The participants were non-randomly assigned to the intervention and control groups due to practical constraints. Specifically, only two therapists were qualified to deliver the NFPP intervention, while the number of families interested in participating exceeded our intervention capacity. To manage this limitation and maintain participant satisfaction, families were assigned to groups based on the order of recruitment:

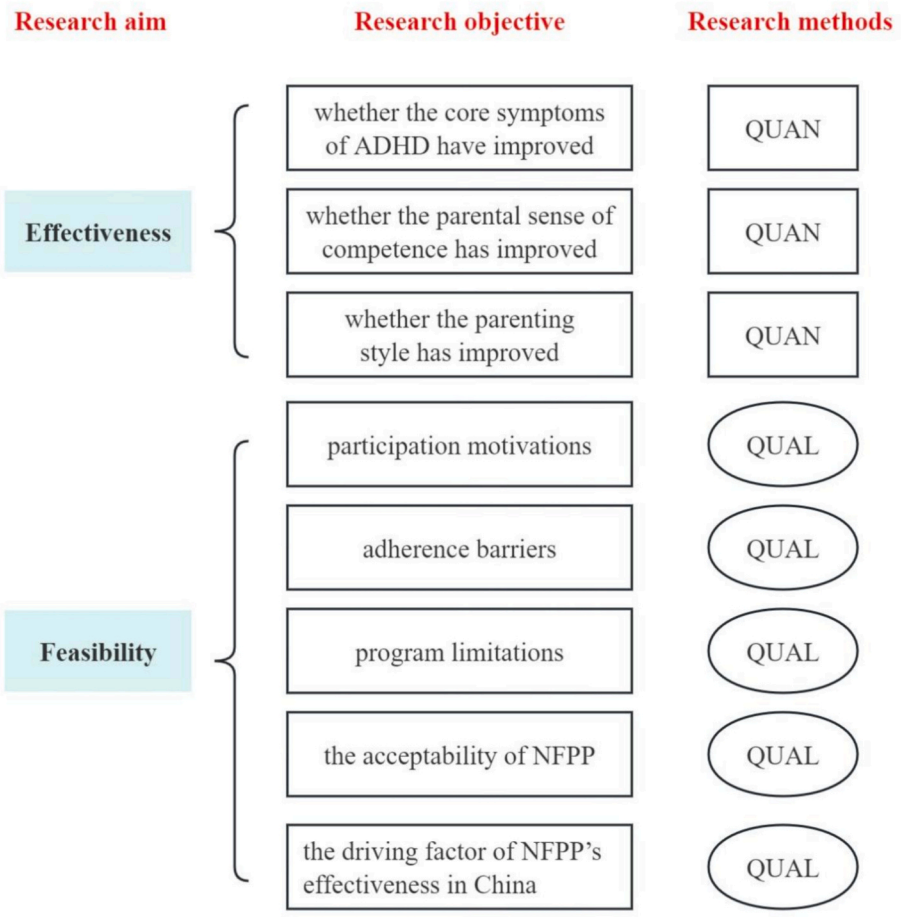


Fig. 1. Research framework.

the first 15 families were assigned to the intervention group to receive the NFPP intervention, while the remaining 15 families were placed on a waitlist and assigned to the control group (scheduled to receive the NFPP intervention six months later). The intervention and control groups were assessed on ADHD core symptoms, parenting styles, and parenting sense of competence both at baseline and post-intervention (week 8). In addition, semi-structured interviews were conducted with the 15 parents in the intervention group. It should be noted that, due to limitations in the study design, the recruitment and data collection for the control group were conducted after a certain delay compared to the intervention group. In addition, during the implementation of the study, participants completed their assessments in separate rooms to prevent sharing of information between the two groups, thereby ensuring data independence. Fig. 2 shows the flowchart of subject recruitment, selection, and study process.

2.3. Intervention

The NFPP is a parent training program developed for parents of children with ADHD. In our study, the NFPP intervention was conducted by two experienced therapists who had received standardized training, supervision, and assessment from the original NFPP team in the UK. Each therapist worked individually with one ADHD family, providing NFPP sessions once a week for eight consecutive weeks, with each session lasting approximately 1.5 h. All sessions were conducted in a comfortable, purpose-designed room, specifically arranged for NFPP implementation. The NFPP program mainly includes ADHD health education, behavior management techniques, scaffolding support, parent-child interaction, and strategies for predicting future behavioral and emotional problems in children (DC et al., 2014). During the 8-week intervention period, parents applied the strategies and skills learned in the sessions to their own children. Supplementary Table 1 presents the weekly content of the NFPP intervention. These are designed to help develop abilities in ADHD children such as sitting still, focusing, taking turns, waiting, negotiating, and compromising.

2.4. Outcome measures

All measures used in this study were based on parent-report. The intervention and control groups were evaluated using the following scales at baseline and after 8 weeks of NFPP: the Chinese Version of Swanson Nolan and Pelham, Version IV Scale-parent form (SNAP-IV) (Swanson et al., 2001; Zhou et al., 2013)was employed to assess the core symptoms improvement of children diagnosed with ADHD, which has satisfactory measurement properties regarding test-retest reliability and validity. Parenting efficacy and satisfaction were assessed using the Chinese Parenting Sense of Competence (C-PSOC) (Ngai et al., 2007) and parenting style was measured with the Parenting Styles Dimensions Questionnaire (PSDQ) (Alagla et al., 2019). Li (Li et al., 2021) and Fu (Fu et al., 2013) conducted reliability and validity analyses of the C-PSOC and PSDQ, and the findings showed that both instruments exhibited satisfactory psychometric properties. All of the above scales were administered by the same researcher.

The primary outcome measure in this study was the SNAP-IV scale. The C-PSOC and PSDQ were included as secondary outcome measures.

2.5. Statistical analysis

2.5.1. Quantitative

The data were analyzed using IBM SPSS 26.0 software. The Shapiro-

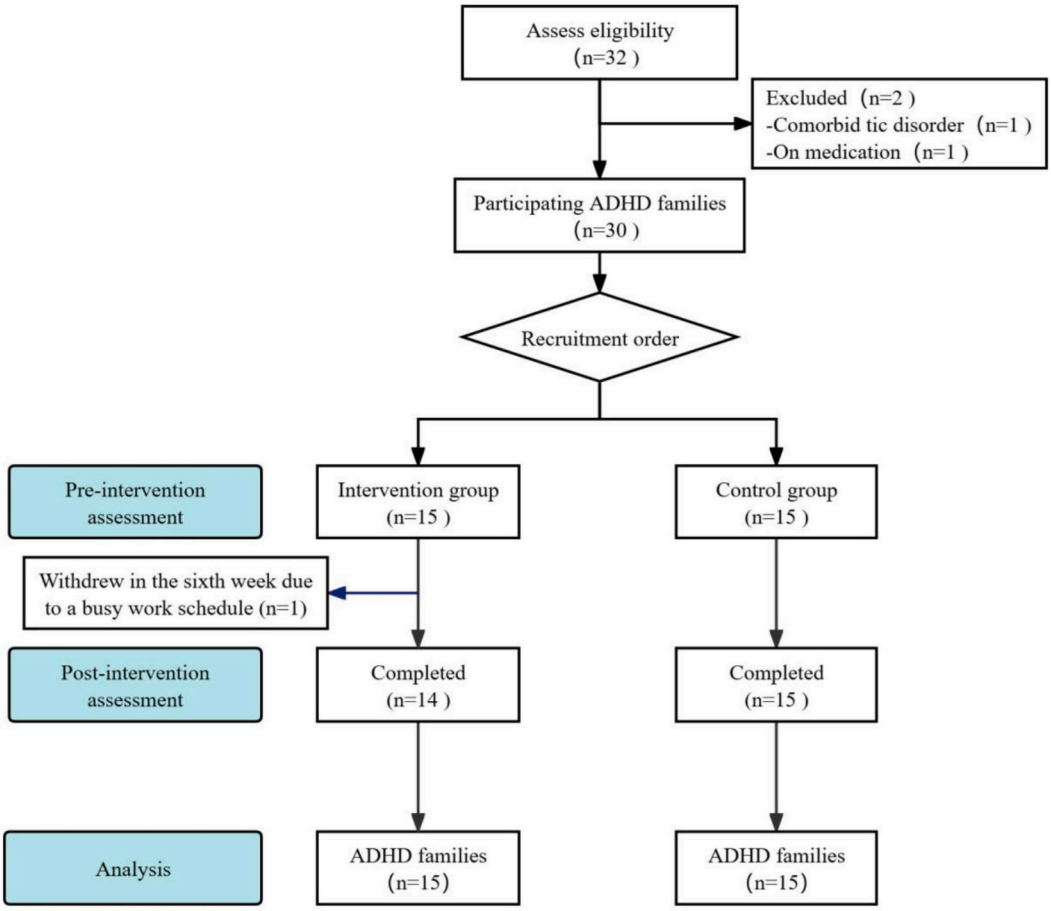


Fig. 2. Participant flowchart.

Wilk test and homogeneity of variance test were performed to assess data normality and variance homogeneity, respectively. Fisher's Exact Test, Chi-square test, and independent samples *t*-test were used to compare age, gender, and education level between the two groups. A two-tailed $P < 0.05$ was considered statistically significant.

Quantitative data with a normal distribution were described as mean ± SD, and group comparisons were performed using independent-samples *t*-tests. Analysis of covariance (ANCOVA) was performed to compare post-intervention scores on the SNAP-IV, PSDQ, and C-PSOC between the intervention and control groups, adjusting for baseline or pre-intervention scores as covariates. To control for the inflation of Type I error associated with multiple comparisons, Bonferroni correction was applied separately to each scale. Cohen's d of 0.8 indicates a "large" effect, while a Cohen's d of 1.2 indicates a "very large" effect (Sawilowsky, 2009). For quantitative data that were not normally distributed, results were summarized as the median [P_{25}, P_{75}]. Between-group comparisons were performed using the Mann–Whitney *U* test, and effect sizes were reported as |r|.

A total of 30 families participated in this study. Among the 15 families undergoing the NFPP intervention, one parent had to withdraw from the intervention two weeks early due to a change in her work schedule. Following the intention-to-treat principle, we included her in the original intervention group for analysis, applying last observation carried forward by using the sixth-week data as a substitute for the missing data from the end of the 8-week intervention.

2.5.2. Qualitative

Two trained researchers (L. Shi and B. Li) conducted individual, face-to-face semi-structured interviews with parents who completed the NFPP intervention (intervention group). All 15 parents were interviewed individually. The interview guide included seven open-ended questions and three structured questions (Tarver et al., 2022). With parental consent, all interviews were audio-recorded and transcribed verbatim. The duration of the interviews ranged from 26 to 40 min (mean [SD], 32.40[4.31]), and all interviews were conducted in a comfortable, private room.

Two researchers independently conducted thematic analysis of the qualitative data using NVivo 12 software. They continuously discussed and revised the coding results to reach agreement. Thematic analysis, a fundamental method for qualitative research, is widely used in qualitative research. It seeks to uncover key themes and patterns within the data to gain a profound understanding. This method allows for the examination of both inductive themes (emerging from the data) and deductive themes (derived from existing theories), offering flexibility in identifying and exploring central themes relevant to the research question, while capturing naturally occurring themes within the dataset (Clarke & Braun, 2013). The analysis process followed the six-stage approach to thematic analysis proposed by Braun & Clarke (Braun & Clarke, 2006).

Data collection was guided by the principle of data saturation, indicating that information from all interviews was adequate. This allowed the generation of rich, nuanced, and multifaceted insights into the phenomenon under investigation. (Ruan et al., 2025). The research team held multiple rounds of in-depth discussions and deliberations on various themes during group meetings to ensure that the findings reflected a collective perspective rather than that of a single researcher, thereby strengthening the rigor and credibility of the analysis.

3. Results

3.1. Quantitative analysis

A total of 30 ADHD children and 30 parents participated in this study (Table 1). There were no statistically significant differences between the intervention and control groups in demographic characteristics of the children and parents. Participants who attended all eight sessions were considered to have completed the intervention. Among the 15 families participating in the NFPP intervention, 14 successfully completed it, resulting in a completion rate of 93.3%.

At baseline, no significant differences were observed between the intervention and control groups on any parent-report scale measures (all $P > .05$). After completing the 8-week NFPP intervention, according to parent-report outcomes, the intervention group showed significantly lower scores for inattention ($F = 12.593$; $P = .004$; $d = 0.647$), hyperactivity/impulsivity ($F = 12.200$; $P = .005$; $d = 0.686$), and oppositional defiant ($F = 7.011$; $P = .040$; $d = 0.880$) compared with the control group.

In addition, compared with the control group, parents in the intervention group reported higher levels of parenting efficacy ($F = 9.263$; $P = .010$; $d = 0.874$) and parenting satisfaction ($F = 7.149$; $P = .025$; $d = 0.950$) after the NFPP intervention, along with improvements in parenting style — characterized by increase authoritative parenting ($F = 12.046$; $P = .005$; $d = 1.058$) and decreased authoritarian ($F = 20.543$; $P < 0.001$; $d = 1.122$) and permissive parenting styles ($F = 9.264$; $P = .015$; $d = 0.863$). These results are presented in Tables 2–4.

Table 2
Comparison of Post-NFPP SNAP-IV Scores Between the Intervention and Control Groups.

	Intervention group ($n = 15$)	Control group ($n = 15$)	F/t	P	Cohen's d
Pre-intervention Inattention, mean (SD)	1.81(0.64)	1.58(0.76)	−0.892	0.389	0.327
Pre-intervention Hyperactivity /impulsivity, mean (SD)	1.32(0.62)	1.24(0.70)	−0.337	0.723	0.121
Pre-intervention Oppositional defiant, mean (SD)	1.38(0.73)	1.38(0.50)	−0.023	0.227	0.009
Post-intervention Inattention, mean (SD)	1.25(0.37)	1.60(0.67)	12.593	0.004	0.647
Post-intervention Hyperactivity /impulsivity, mean (SD)	0.87(0.40)	1.22(0.60)	12.200	0.005	0.686
Post-intervention Oppositional defiant, mean (SD)	0.83(0.43)	1.31(0.64)	7.011	0.040	0.880

Note. ANCOVA is used to adjust for baseline or pre-NFPP scores. P values for post-intervention comparisons were adjusted using the Bonferroni correction to control for multiple comparisons within each scale. Statistical significant difference for $P < .05$.

3.2. Qualitative analysis

A total of 15 parents participated in semi-structured interviews, including 2 males (13.3%) and 13 females (86.7%). The interviews identified six themes: (a) motivation, (b) knowledge and theoretical parenting skills acquisition, (c) understanding and respect: reshaping parental beliefs, (d) active participation in parenting practices, (e) barriers, and (f) limitations. Each theme was further subdivided into several subthemes as follows: (a) *teacher complaints, concerns about the child's future development, and difficulty in management*; (b) *illness-related knowledge, parenting skills-related knowledge*; (c) *capacity development, parenting styles*; (d) *communication and sharing, changing parenting styles, and putting knowledge into practice*; (e) *time cost, spousal involvement*; (f) *hope for ongoing follow-up in the future, literal translation of some terms*. The themes, subthemes, and their frequencies are summarized in Supplementary Table 2, while Table 5 presents the themes and subthemes along with representative supporting quotations.

3.2.1. Theme 1—Motivation

When asked why they participated in the NFPP program, parents expressed concerns about their children's difficulties in school, at home, and in their daily lives. This theme has been organized into three subthemes: teacher complaints, concerns about the child's future development, and difficulties in behavior management.

3.2.1.1. Teacher complaints. Parents reported that the primary reason for participating in NFPP stemmed from teachers' complaints regarding classroom discipline, emotional management, and academic progress of their children. Observations of unusual behaviors by some professionally trained teachers also helped identify potential symptom and encouraged parents to seek medical help.

3.2.1.2. Concerns about the child's future development. Furthermore, parents aimed to effectively support their children's development, hoping to see improvements in their abilities to face challenges at school and in life. And they hoped to see their children grow into better individuals.

3.2.1.3. Difficulty in management. These children often experienced emotional dysregulation, learning difficulties, and social challenges. Undoubtedly, these issues undoubtedly increased parental stress and anxiety, which also diminished their confidence and satisfaction in parenting to some extent.

Table 1
Demographic Characteristics of the Participants.

Characteristic	Participants, No. (%) ($N = 60$)					
	Children (n = 30)			Parents($n = 30$)		
	Intervention group (n = 15)	Control group ($n = 15$)	P	Intervention group (n = 15)	Control group (n = 15)	P
Sex			1.000			1.000
Male	14 (93.3)	13 (86.7)		2 (13.3)	2 (13.3)	
Female	1 (6.7)	2 (13.3)		13 (86.7)	13 (86.7)	
Age, mean (SD), y	7.53 (1.51)	8.13 (1.64)	0.306	32.47 (1.77)	33.67 (1.80)	0.076
Education			1.000			0.269
Preprimary education	1 (6.7)	2 (13.3)				
Primary education	14 (93.3)	13 (86.7)				
Higher professional education				8 (53.3)	5 (33.3)	
University education or above				7 (46.7)	10 (66.7)	

Table 3
Comparison of Post-NFPP C-PSOC Scores Between the Intervention and Control Groups.

	Intervention group ($n = 15$)	Control group ($n = 15$)	F/t/Z	*P*	Cohen's d/|r|
Pre-intervention Efficacy, mean (SD)	27.33(5.80)	27.13(6.08)	−0.092	0.855	0.034
Pre-intervention Satisfaction, median (P_{25}, P_{75})	34.00(32.00,36.00)	33.00(30.00,37.00)	−0.146	0.884	0.027
Post-intervention Efficacy, mean (SD)	33.33(3.24)	29.13(5.97)	9.263	0.010	0.874
Post-intervention Satisfaction, mean (SD)	36.33(5.31)	31.40(5.07)	7.149	0.025	0.950

Note. ANCOVA is used to adjust for baseline or pre-NFPP scores. P values for post-intervention comparisons were adjusted using the Bonferroni correction to control for multiple comparisons within each scale. Statistical significant difference for $P < .05$. Cohen's d and |r| represent the effect sizes for *t*-tests and Mann–Whitney *U*- tests, respectively.

Table 4
Comparison of Post-NFPP PSDQ Scores Between the Intervention and Control Groups.

	Intervention group ($n = 15$)	Control group ($n = 15$)	F/t/Z	*P*	Cohen's d/|r|
Pre-intervention PSDQ Self-rating					
Authoritative parenting style, mean (SD)	90.27(12.30)	88.60(13.00)	−0.361	0.556	0.132
Authoritarian parenting style, mean (SD)	47.80(11.33)	47.67(11.84)	−0.032	0.884	0.011
Permissive parenting style, median (P_{25}, P_{75})	38.00(36.00,40.00)	34.00(32.00,42.00)	−0.665	0.506	0.121
Pre-intervention PSDQ Other-rating					
Authoritative parenting style, mean (SD)	82.47(13.90)	81.73(13.22)	−0.148	0.692	0.055
Authoritarian parenting style, median (P_{25}, P_{75})	42.00(39.00,48.00)	45.00(34.00,54.00)	−0.187	0.852	0.034
Permissive parenting style, mean (SD)	36.87(4.42)	37.73(5.90)	0.455	0.240	0.165
Post-intervention PSDQ Self-rating					
Authoritative parenting style, mean (SD)	102.33(14.30)	88.40(11.93)	12.046	0.005	1.058
Authoritarian parenting style, mean (SD)	37.13(6.27)	46.60(10.15)	20.543	<0.001	1.122
Permissive parenting style, mean (SD)	31.33(6.16)	36.47(5.74)	9.264	0.015	0.863
Post-intervention PSDQ Other-rating					
Authoritative parenting style, mean (SD)	99.87(12.41)	82.27(11.78)	49.120	<0.001	1.455
Authoritarian parenting style, mean (SD)	37.93(8.98)	45.07(10.94)	10.790	0.008	0.713
Permissive parenting style, mean (SD)	31.60(4.81)	37.47(5.67)	13.618	0.003	1.117

Note. ANCOVA is used to adjust for baseline or pre-NFPP scores. P values for post-intervention comparisons were adjusted using the Bonferroni correction to control for multiple comparisons within each scale. Cohen's d and |r| represent the effect sizes for *t*-tests and Mann–Whitney *U*- tests, respectively. Statistical significant difference for $P < .05$.

3.2.2. Theme 2—Knowledge and theoretical parenting skills acquisition

At the beginning of the NFPP program, parents received a comprehensive introduction to ADHD, as gaining knowledge about the disorder is often the first step to effective intervention. In the interviews, parents reported significant improvement in their understanding of ADHD characteristics and parenting skills.

3.2.2.1. Illness-related knowledge. All parents reported that they previously had little understanding of ADHD. Throughout the intervention, they gradually developed a deeper understanding of the causes, symptoms, and functional impairments associated with ADHD.

3.2.2.2. Parenting skills–related knowledge. As the intervention progressed, parents acquired tailored parenting skills for ADHD to support the development of their children's emotional regulation, organization, and time perception abilities.

3.2.3. Theme 3—Understanding and respect: Reshaping parental beliefs

The NFPP provided participants with ADHD-related theoretical knowledge at the beginning to enhance the overall understanding of the disorder. Based on this foundation, parents' perspectives on their children's issues were changing.

3.2.3.1. Capacity development. Additionally, several parents mentioned the importance of setting appropriate goals and tasks for their children based on their developmental level, rather than solely assessing their abilities based on academic performance. This also reflected parents' comprehensive acceptance of their children, offering them more understanding and compassion.

3.2.3.2. Parenting styles. Moreover, changing parenting styles is crucial, as they profoundly impact children's developmental processes. Parents noted a positive shift in their attitudes toward their children, no longer resorting to simple punishment or neglecting issues but attempting to interact with them more patiently, showing more attention to their emotional needs.

3.2.4. Theme 4—Active participation in parenting practices

Parents actively engaged in parenting practices and most parents integrated them into daily lives. This theme is divided into three subthemes: communication and sharing, changing parenting styles, and putting knowledge into practice.

3.2.4.1. Communication and sharing. Throughout the intervention process, the support of partners and family members was crucial, with almost all parents mentioning sharing their NFPP-acquired knowledge with family, friends, or colleagues, and exchanging parenting experiences. This facilitated parenting consistency and alleviated individual parenting pressures.

3.2.4.2. Changing parenting styles. Additionally, the most obvious change mentioned by parents was the shift in their parenting approach. They ceased resorting to scolding or reacting negatively, but instead strove to understand, respect, and support their children, fostering a harmonious parent-child atmosphere.

Table 5
Qualitative Themes, Subthemes, and Supporting Quotations.

Themes	Subthemes	Supporting Quotations
Motivation	Teacher complaints	*"Overall, the teacher's feedback was not positive, noting issues such as talking with classmates and lack of focus during class." (Parent 5)* *"The teacher also mentioned that our child often gets into conflicts with classmates at school, cannot sit still in class, and frequently seems distracted. The teacher also noted that he seems somewhat different from others and suggested that we consult a doctor "(Parent 6)*
	Concerns about the child's future development	*"... It's more about his future, his ability to adapt to life and society…it's more about addressing the current issues and hoping for overall improvement in the child." (Parent 12)*
	Difficulty in management	*"He often loses his temper, unwilling to admit his own mistakes, especially when it comes to homework errors. He cries almost every two to three days, having a big emotional outburst. "(Parent 8)* *"Managing our child's behavior and emotions has proven quite challenging. Unexpected situations frequently arise, leaving us uncertain about how to address them. "(Parent 13)*
Knowledge and theoretical parenting skills acquisition	Illness-related knowledge	*"Another point is understanding that these children can't wait, get angry easily, and procrastinate not because they mean to, but because this is what ADHD kids are like. "(Parent 12)* *"We're no longer like before, always thinking he's being difficult, disobedient, or acting out on purpose. Now we understand what challenges he's actually facing…" (Parent 8)*
	Parenting skills–related knowledge	*"We didn't know much about behavior management before. If he misbehaved, we would often resort to scolding or spanking. It was only after taking this course that we realized there are effective techniques for interacting with children. Now, we've mastered many communication techniques and strategies for improving behavior, which has been really helpful for us." (Parent 11)* *"So, it turns out that talking to children requires some skills. "(Parent 4)*
Understanding and respect: reshaping parental beliefs	Capacity development	*"Now I can better understand my child and realize the importance of setting goals that are suitable for him, rather than setting excessively high expectations. "(Parent 15)* *"In the past, I used to expect my child to meet the traditional expectations of a typical Chinese child. Now I realize that there are some things he cannot do, which are beyond his capabilities. "(Parent 3)*
	Parenting styles	*"...Um, there's also a change in the way we understand and approach education and discipline methods, um, um… we shouldn't turn parent-child relationships into purely negative management. It's just a change in mindset. "(Parent 3)* *"Now that I know, I can choose a different approach to parenting, seldom engaging in direct confrontation with my child. "(Parent 10)*
Active participation in parenting practices	Communication and sharing	*"Sometimes, we communicate with each other about whether our strategies are correct and whether they contradict parenting manuals. Occasionally, we remind each other of areas where we could improve. "(Parent 5)* *"We held family meetings to share and monitor how we were implementing the strategies together, and we also shared them with our friends. "(Parent 12)*
	Changing parenting styles.	*"Changing my way of speaking, I've reduced scolding and punishment toward my child. "(Parent 7)* *"I speak to him gently but firmly, expressing my perspective without judgment, and my tone is pleasant." (Parent 14)* *"...he's willing to spend time with his child, understand their behavior, and has stopped hitting them. "(Parent 4)*
	Putting knowledge into practice	*"The main thing is that I absorb the information and combine it with my own situation. I immediately apply what I've recently learned. For methods or strategies that I don't understand yet, I note them down and find opportunities to implement them later. "(Parent 4)*
Barriers	Time cost	*"There are definitely obstacles in terms of time, like for me, I only have time to be with her in the evenings. "(Parent 3)* *"The time spent with our child is too short. Both of us are extremely busy with work, making it challenging to dedicate enough time to implementing the strategies we've learned. "(Parent 13)* *"The younger brother comes over to interrupt us because at nine o'clock, he has to go find Mom. So whatever we're doing gets disrupted, which can affect our focus and learning outcomes." (Parent 7)* *"Another issue is my second child. Not having anyone to accompany him has caused me considerable difficulty. "(Parent 9)*
	Spousal involvement	*"...Dad felt he needed to read to finish the homework for the day, but I thought it was enough… It was already 11 o'clock, so I told him to go back to bed, but Dad*

(continued on next page)

Table 5 (*continued*)

Themes	Subthemes	Supporting Quotations
		insisted he had to finish reading before going to sleep. "(Parent 12)
		"...When we encounter problems, the two of us often struggle to reach a consensus. "(Parent 8)
		"The main thing is that Dad doesn't learn this stuff, and later he told me he wants to quit because he doesn't have enough energy." (Parent 2)
		"I told him that I needed to hold a family meeting, and he just said, "Then go ahead and do it." Just that one sentence, and I couldn't continue the conversation. "(Parent 1)
Limitations	Hope for ongoing follow-up in the future	*"…and I'm worried that it will be difficult to seek help after the course is over, I'm a little anxious about this and hope to have ongoing support in the future. "(Parent 6)*
		"The subsequent challenges, assistance, support, and so on, we hope to continuously seek this kind of help in the future, that is, to regularly guide us on what we're doing wrong and how to improve. "(Parent 7)
		"Could it be that there are some follow-up visits, just to strengthen it, not necessarily to sit down… It could be done through an assistant… Like making a phone call… Just reminding us at some point… Maybe then we'll remember again. "(Parent 10)
	Literal translation of some terms	*"I still feel that the translation is too literal… For example, some terms like 'scaffolding' and 'delay' sound awkward. It took me a long time to figure out what 'scaffolding support' even meant. "(Parent 12)*

3.2.4.3. Putting knowledge into practice. Parents learned coping strategies and techniques through NFPP, enabling them to approach their children with greater confidence and apply these strategies in their daily routines. They continuously reflected on experiences, leading to improvements in their children's behavior throughout this process.

3.2.5. Theme 5—Barriers

Obstacles parents encountered during the implementation of strategies would reduce their execution and impact the effectiveness of the intervention.

3.2.5.1. Time cost. The lack of available time for implementing strategies was reported as the biggest issue by parents. They acknowledged being frequently occupied with work, which indeed posed significant difficulties. Parents expressed a desire for dedicated one-to-one time with their child with ADHD, but interruptions from siblings frequently occurred. Consequently, it became difficult for parents to dedicate their full attention and energy to the child with ADHD.

3.2.5.2. Spousal involvement. Parents' comprehension of the curriculum content was important, as it influenced their strategy choices in specific situations. Another obstacle was the frequent disagreement between parents in using different strategies to deal with their children's behavioral and emotional issues, making it difficult at times to reach a consensus on motivating the children to exhibit the expected behavior. Additionally, fathers seemed to show less patience than mothers in parenting matters.

3.2.6. Theme 6—Limitations

Although parents generally expressed positive attitudes toward the NFPP and acknowledged its benefits, they also articulated their expectations for ongoing support and program improvement.

3.2.6.1. Hope for ongoing follow-up in the future. Many parents strongly hoped to maintain long-term contact with the therapists after the course ended, believing that continued follow-up would help reinforce what they had learned and sustain positive parenting behaviors. Although the NFPP enhanced parents' confidence and skills, they still felt a need for continuous guidance to maintain behavioral change. This expectation for ongoing support also revealed an active learning attitude—parents no longer regarded themselves as passive recipients of knowledge but as active participants seeking continuous improvement.

3.2.6.2. Literal translation of some terms. Some noted that certain terms translated from English into Chinese were occasionally confusing, although therapists' explanations could partly resolve this issue. These reflections underscored the need for cultural and linguistic adaptation of NFPP to ensure that parents fully comprehend and effectively apply its strategies. Beyond practical concerns, such reflections revealed a process of meaning-making—parents actively interpreted program concepts and sought to integrate them within their own cultural and familial contexts.

3.2.7. Structured questions

All parents indicated that they would recommend NFPP to friends, expressed a desire for it to be a part of their usual care from their clinician, and believed that the intervention should be routinely offered to parents seeking care for their child with ADHD.

3.3. Comparison of quantitative and qualitative findings

The quantitative and qualitative findings were generally consistent and mutually reinforcing. The quantitative analysis showed that after the 8-week NFPP intervention, parents reported significant reductions in children's inattention, hyperactivity/impulsivity, and oppositional behaviors, along with notable improvements in parenting efficacy, satisfaction, and authoritative parenting style.

In the semi-structured interviews, many parents described the positive impact of the NFPP on both their children and families. They reported reduced time spent on homework, fewer classroom restlessness behaviors, fewer complaints from teachers, and enhanced competence in managing children's behavioral problems. Consequently, many parents expressed a willingness to recommend the NFPP to other families, further supporting the program's effectiveness and acceptability from a subjective perspective.

Moreover, quantitative data indicated an increase in authoritative parenting style. Qualitative findings supported this by showing that parents regarded a better understanding of ADHD as one of the most valuable outcomes of NFPP participation. This deeper understanding enhanced their overall cognitive insight and contributed to a more empathetic and tolerant attitude toward their children. Parents consistently applied the parenting skills learned during the program in their daily routines, which reduced the use of harsh discipline, improved their ability to cope with the challenges of raising a child with ADHD, and ultimately facilitated a shift toward a more authoritative and supportive parenting style.

In addition, parents reported that learning concrete parenting strategies and observing positive responses from their children helped them

establish a closer emotional connection. These rewarding experiences enhanced their sense of confidence in parenting, leading to greater satisfaction and fulfillment in parent–child interactions.

4. Discussion

This study examined the effectiveness and feasibility of the NFPP among families of children with ADHD in China. The results indicate that the NFPP effectively improves parent-report core ADHD symptoms, and both the attendance rate and semi-structured interview findings suggest that NFPP is feasible and well-accepted in the Chinese context.

This study reveals that NFPP is effective for Chinese children with ADHD. Compared with the control group, the intervention group showed significant reductions in parent-report inattention, hyperactivity/impulsivity, and oppositional defiant scores after the NFPP intervention. In addition, parents demonstrated improved parenting efficacy and satisfaction, along with an increased use of an authoritative parenting style. These findings are consistent with those reported by Daley et al., who also found that the NFPP improved ADHD symptoms while enhancing parenting efficacy and satisfaction (Daley & O'Brien, 2013). Sonuga-Barke et al. further compared the effectiveness of NFPP with treatment-as-usual and found that NFPP produced notable improvements in oppositional defiant behaviors but only minimal effects on inattention and hyperactivity/impulsivity. A plausible explanation is that the study adopted broad inclusion and exclusion criteria to maximize recruitment of children with ADHD, without applying strict diagnostic standards. As a result, many children presented with comorbid difficulties and developmental delays, which may have led to an underestimation of NFPP's true therapeutic effects (Sonuga-Barke et al., 2018). Deng et al. (Deng et al., 2024) also conducted a quantitative study on the effectiveness of the NFPP among Chinese children with ADHD. Although their findings suggested that NFPP significantly improved core ADHD symptoms, the control group in their study received routine health education, and the hyperactivity and impulsivity scores of the control group also showed significant improvement after the intervention, indicating that health education itself might have contributed to certain behavioral improvements. Therefore, the study may not have accurately isolated the specific effects of NFPP.

Compared with other nonpharmacological interventions implemented in China, the NFPP appears to offer broader benefits for both parents and children. The MBSR program delivered to parents of children with ADHD in Hong Kong, as reported by Law et al., showed that MBSR can effectively reduce parental stress; however, it had no substantial impact on improving ADHD symptoms in children (Law et al., 2025). One likely reason is that the MBSR intervention focused primarily on cultivating awareness of emotions, thoughts, and bodily sensations to enhance parental well-being, but did not deeply address child-related issues or parenting strategies. Consequently, the benefits for children with ADHD were limited. In contrast, NFPP places a stronger focus on improving parenting styles, strengthening positive parenting behaviors, and promoting constructive parent–child interactions. These changes help improve the parent–child relationship and lead to more effective reductions in ADHD-related behaviors. Importantly, improvements in parenting behaviors function as a key mediator linking the intervention to changes in child behavior, thereby indirectly contributing to the meaningful behavioral improvements observed in children (R et al., 2014). Consequently, when compared with other non-pharmacological interventions conducted in Chinese cultural contexts, NFPP may provide more comprehensive and targeted support for both parents and children.

Among the 15 families in the intervention group, all completed the baseline assessment (100%), 14 families completed the eight-week NFPP intervention (93.3%), and all 15 families completed both the post-intervention assessment (100%) and the semi-structured interviews (100%). Parents in the intervention group showed strong motivation to participate, mainly due to concerns raised by teachers and worries about their children's future. They hoped that NFPP could help improve their child's behavior and enhance the parent–child relationship. Despite obstacles such as busy work schedules, overall adherence remained high, with 93.3% of families completing all sessions. Parents reported that they learned useful knowledge and skills, became more understanding and respectful toward their children, and actively applied parenting strategies in daily life. Although minor challenges related to the Chinese translation of NFPP materials existed, the therapists' explanations mitigated these effects. Furthermore, all parents indicated that they would recommend NFPP to friends, expressed a desire for it to be part of routine clinical care, and believed that the intervention should be regularly offered to parents of children with ADHD. Taken together, these findings demonstrate that NFPP shows high acceptability and good feasibility among Chinese families of children with ADHD.

The semi-structured interviews indicated that Chinese parents' motivation to participate in the NFPP was primarily driven by teacher complaints, difficulties in managing their child's behavior, and concerns about the child's future development. In contrast, Daley's qualitative study of families in the United Kingdom identified a different pattern: UK parents mainly joined the NFPP to learn more knowledge. Different motivations may stem from cultural differences between East and West. Influenced by traditional Confucianism, Chinese parents prioritize social norms, interpersonal harmony, and the importance of hard work (du Prel Carroll et al., 2012; Fu et al., 2022). Additionally, academic achievement remains highly valued. Therefore, Chinese parents prioritize evaluation metrics that may influence their children's future development, ultimately driving their healthcare-seeking behavior.

In the interviews, parents consistently reported gaining a deeper understanding of ADHD and relevant parenting strategies, suggesting a meaningful improvement in their knowledge. With this enhanced understanding, many parents began to reflect on the underlying causes of their children's behavioral difficulties. Their attitudes gradually shifted from blame and punishment toward greater understanding, acceptance, and support. They also reported applying the strategies learned in NFPP to everyday parenting, reducing the use of harsh discipline and adopting more positive approaches. These changes, in turn, appeared to contribute to improvements in their children's ADHD symptoms. Previous research has shown that emotional maltreatment, as well as negative, overprotective, or overly controlling parenting practices, are risk factors for ADHD, whereas NFPP emphasizes enhancing parent–child relationships and improving child behavior through promoting positive parenting strategies (Gau, 2007; Johnston, 1996). The KAP theory, proposed by Dr. Gcust in the 1960s, posits that individuals first acquire information about specific behaviors, then undergo attitude changes toward these behaviors, resulting in behaviors consistent with these attitudes (Bettinghaus, 1986). Thus, the KAP framework may partially explain the patterns observed in our study: improvements in parental knowledge and attitudes may have facilitated positive changes in parenting behavior, thereby contributing to symptom improvement in children. However, it should be noted that although parents reported improved understanding in the interviews, this study did not quantify parental knowledge levels before the intervention. Therefore, the role of the KAP model in explaining the mechanisms of NFPP requires further evaluation in future research.

Parents' and caregivers' understanding of the disorder plays a crucial role in the primary healthcare of children and adolescents (Gajria et al., 2014; Khan & Aslani, 2020). Insufficient parental understanding of ADHD may lead to delays in identification and treatment (Park et al., 2018). The European ADHD Guidelines Group (EAGG) suggests that most treatment goals for ADHD can be achieved by providing high-quality psychoeducation at the outset of treatment, helping families better understand ADHD and its management, and considering their concerns, attitudes, and beliefs about ADHD (Coghill et al., 2023). Additionally, parents' understanding of their children's challenging behaviors determines their treatment needs (Bussing et al., 2012; dosReis et al., 2007); therefore, we emphasize the significant role of parental

cognition, as acquiring knowledge is often the first step in ADHD treatment and can significantly influence treatment outcomes. Many parents in China may lack adequate awareness and understanding of ADHD (Li, Wang, et al., 2022). Therefore, we recommend providing high-quality psychoeducation for Chinese ADHD families to enhance parental awareness, thereby promoting positive treatment outcomes.

Although our study demonstrates that NFPP is effective for Chinese children with ADHD, as NFPP originates from the UK, the differences in understanding and cultural contexts between China and the UK may limit some of its therapeutic potential. Therefore, it is crucial to make cultural adaptations to NFPP to better suit the Chinese context. In addition, although parents clearly expressed a need for continued support during the interviews, the design of the present study does not permit an assessment of the long-term impact or sustainability of NFPP. Future research should incorporate randomized controlled trials with extended follow-up periods to determine whether the short-term benefits observed can be maintained over time.

In summary, this study offers preliminary evidence supporting the use of NFPP among Chinese families of children with ADHD, showing meaningful improvements in parent-report child symptoms and parental parenting competence. The findings further indicate that NFPP is both feasible and acceptable in Chinese context. Together, these results suggest that NFPP could serve as a valuable complement to routine clinical services, providing more structured support for families of children with ADHD and offering clinicians a practical, evidence-based framework for delivering parent-training interventions.

5. Limitations

Our study has some limitations. Firstly, this study employed both quantitative and qualitative methods to address two distinct research aims: quantitative data were used to evaluate the effectiveness of NFPP, while qualitative data focused on its feasibility. As a result, the quantitative and qualitative findings could not be integrated into a comprehensive conclusion as traditional mixed-methods research. In the future, we plan to design a rigorous mixed-methods framework to fully leverage the complementary strengths of quantitative and qualitative research, thereby enhancing our understanding of the research questions. Secondly, child behavioral outcomes in the present study were assessed primarily through proximal informant reports, namely parent-report measures. Although parent reports are commonly used and clinically meaningful outcomes in parent training research, they may be influenced by changes in parental perceptions, expectations, or reporting bias following the intervention. Therefore, the observed improvements in child behavior should be interpreted with caution. Future studies should incorporate multiple informants, such as teacher reports, as well as more objective behavioral assessments, to evaluate the consistency of intervention effects across different contexts and reduce potential informant bias. Thirdly, we assessed participants only before and after the intervention, which limits the ability to evaluate the long-term effects and sustainability of NFPP. Future research should incorporate randomized controlled trials with extended follow-up periods to determine whether the short-term benefits of NFPP can be sustained over time. Fourthly, the participants in this study were recruited from hospital outpatient clinics, which may introduce a degree of selection bias. Families actively seeking clinical services often display stronger motivation for intervention and greater concern regarding their children's symptoms. As a result, the current sample may not fully represent the broader population of families of children with ADHD. This motivation-related bias could have positively influenced both intervention engagement and perceived benefits. Future research should consider recruiting participants from community settings, schools, or broader population-based samples to enhance representativeness and external validity. Fifthly, we did not use random methods for group assignment. Although there were no significant differences in baseline characteristics between the two groups, we cannot guarantee that there were no other potential confounding factors affecting the final results. In addition, the delayed recruitment and data collection of the control group may have introduced influences other than the NFPP intervention. Future randomized controlled trials are needed to further validate the conclusions of this study. Finally, the relatively small sample size may limit the generalizability of our findings. To improve the robustness and applicability of future results, we plan to include larger samples in subsequent studies.

6. Conclusion

This non-randomized controlled trial demonstrates the effectiveness of NFPP in reducing parent-report core symptoms of ADHD in children and enhancing parents' parenting competence and use of authoritative parenting styles. Although parents encountered some challenges in implementing the strategies, NFPP was generally considered feasible and acceptable. In addition, some cultural adaptations may be needed to better tailor NFPP to Chinese families.

Abbreviations

ADHD attention-deficit/hyperactivity disorder
NFPP the New Forest Parenting Programme
SNAP-IV the Chinese Version of Swanson Nolan and Pelham, Version IV Scale-parent Form
C-PSOC the Chinese Parenting Sense of Competence
PSDQ the Parenting Styles Dimensions Questionnaire
ANCOVA analysis of covariance
KAP theory the Knowledge, Attitude, and Practice theory

CRediT authorship contribution statement

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Consent for publication

Not applicable.

Ethics approval and consent to participate

The study has received informed consents from all participants, their parents or legal guardians. The study was approved by the Ethics Committee of Jinan University (JNUKY-2022-033). Clinical trial number: not applicable.

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Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.actpsy.2026.106492.

Data availability

The dataset used and analyzed during the current study are available from the corresponding author on reasonable request.

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